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International University in Tashkent

System Analysis & Design

Tutorial 03

Agenda

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1. Previous tutorial recap
    1. Data flow diagram (DFD)
    2. Activity diagram (AD)
2. Use case
3. Use case diagram

Previous tutorial recap

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1. **Data flow diagram:**
    1. What is it?
    2. What shapes does it have?
    3. What does it document?

2. **Activity diagram:**
    1. What is it?
    2. What shapes does it have?
    3. What does it document?

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Use case

Documenting actions between a user and system to achieve a goal.

Use Case: Example

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Use Case Number	ID to represent your use case
Application	What system or application does this pertain to
Use Case Name	The name of your use case, keep it short and sweet
Use Case Description	Elaborate more on the name, in paragraph form.
Primary Actor	Who is the main actor that this use case represents
Precondition	What preconditions must be met before this use case can start
Trigger	What event triggers this use case
Basic Flow	The basic flow should be the events of the use case when everything is perfect; there are no errors, no exceptions. This is the "happy day scenario". The exceptions will be handled in the "Alternate Flows" section.
Extensions (Alternate Flows)	The most significant alternatives and exceptions

Source: http://www.gatherspace.com/static/use_case_example.html

What is Use Case?

Google

“A use case is a specific, goal-oriented scenario describing how a user (or "actor") interacts with a system to achieve a particular outcome, such as placing an order on a website or a customer withdrawing cash from an ATM. In software and systems engineering, use cases define the system's functional requirements from the user's perspective, helping to clarify goals, plan development, and serve as a basis for testing and documentation.”

https://www.google.com/search?q=use+case

Wikipedia

“In both software and systems engineering, a use case is a structured description of a system’s behavior as it responds to requests from external actors, aiming to achieve a specific goal. The term is also used outside software/systems engineering to describe how something can be used.”

Source: https://en.wikipedia.org/wiki/Use_case

Use Case: Example

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Use Case Number	ID to represent your use case
Application	What system or application does this pertain to
Use Case Name	The name of your use case, keep it short and sweet
Use Case Description	Elaborate more on the name, in paragraph form.
Primary Actor	Who is the main actor that this use case represents
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Extensions (Alternate Flows)	The most significant alternatives and exceptions

Source: http://www.gatherspace.com/static/use_case_example.html

Use Case

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As you can see, a “Use case” is just a table of text that documents specific user-system interaction.

It has:

- Id
- Name
- Description,
- Actors
- Steps (happy path)
- Extensions (edge cases)
- Etc.

Let’s talk about each of these in detail.

Use case: Fields

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Use Case Number	ID to uniquely represent your use case
Application*	What system or application does this use case belongs to. For example “Accounting” subsystem or “Mobile API”.
Use Case Name	The name of your use case. Keep it short, but make sure it is descriptive. For example: User registration, Editing blog post.
Use Case Description	Elaborate more on the name, in paragraph form.
Primary Actor	Who is the main actor that this use case represents?
Precondition*	What preconditions must be met before this use case can start? For example: User registrations are enabled, post belongs to the user, etc.
Postconditions*	What will be the guaranteed outcomes of this use case? For example: User data inserted into the database, logged in and redirected to the homepage.
Trigger	What event triggers this use case? For example: User clicks on register button in the registration form.
Basic Flow	The basic flow should be the events of the use case when everything is perfect; there are no errors, no exceptions. This is the "happy path scenario". The exceptions will be handled in the “Extensions (Alternate Flows)" section.
Extensions (Alternate Flows)	The most significant alternatives and exceptions.

Adopted from: http://www.gatherspace.com/static/use_case_example.html

Exercise: Group

1. Open “SAD-Tutorial-02-Cases.pdf” file.
2. Read the first case.
3. Create the Use Case for it on the whiteboard.

Exercise: Individual

1. Take 15 minutes to read and create 2 more Use Cases (on your own).

2. Anyone would like to draw their version on the whiteboard?

3. Discuss how your Use case is similar or different from the one on the whiteboard.

Note: You can create it on a piece of paper or use any online tool.

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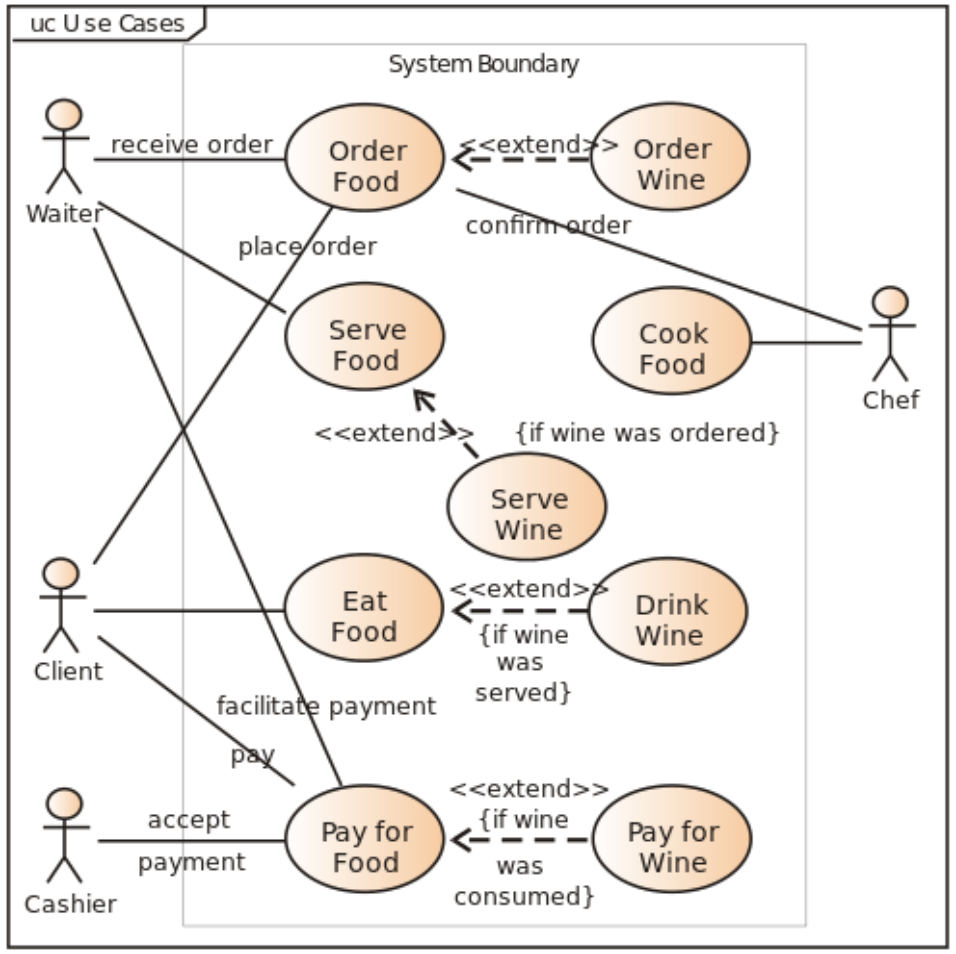
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Use Case Diagram

Visual depiction of how users may interact with the system.

Use Case Diagram: Example

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By Kishorekumar 62, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=7880320

What is Use Case?

Google

"A use case diagram is a high-level, visual representation of a system's intended functionality, showing the interactions between users (called actors) and the various functions (use cases) the system offers. It helps define the system's scope and major features from a user's perspective, serving as a communication tool for understanding how the system will be used without detailing its internal operations."

https://www.google.com/search?q=use+case+diagram

Wikipedia

"A use case diagram is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The use cases are represented by either circles or ellipses. The actors are often shown as stick figures."

Source: https://en.wikipedia.org/wiki/Use_case_diagram

Use Case Diagram: Shapes

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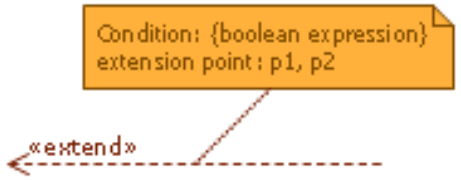
Communication path



Include



Extend

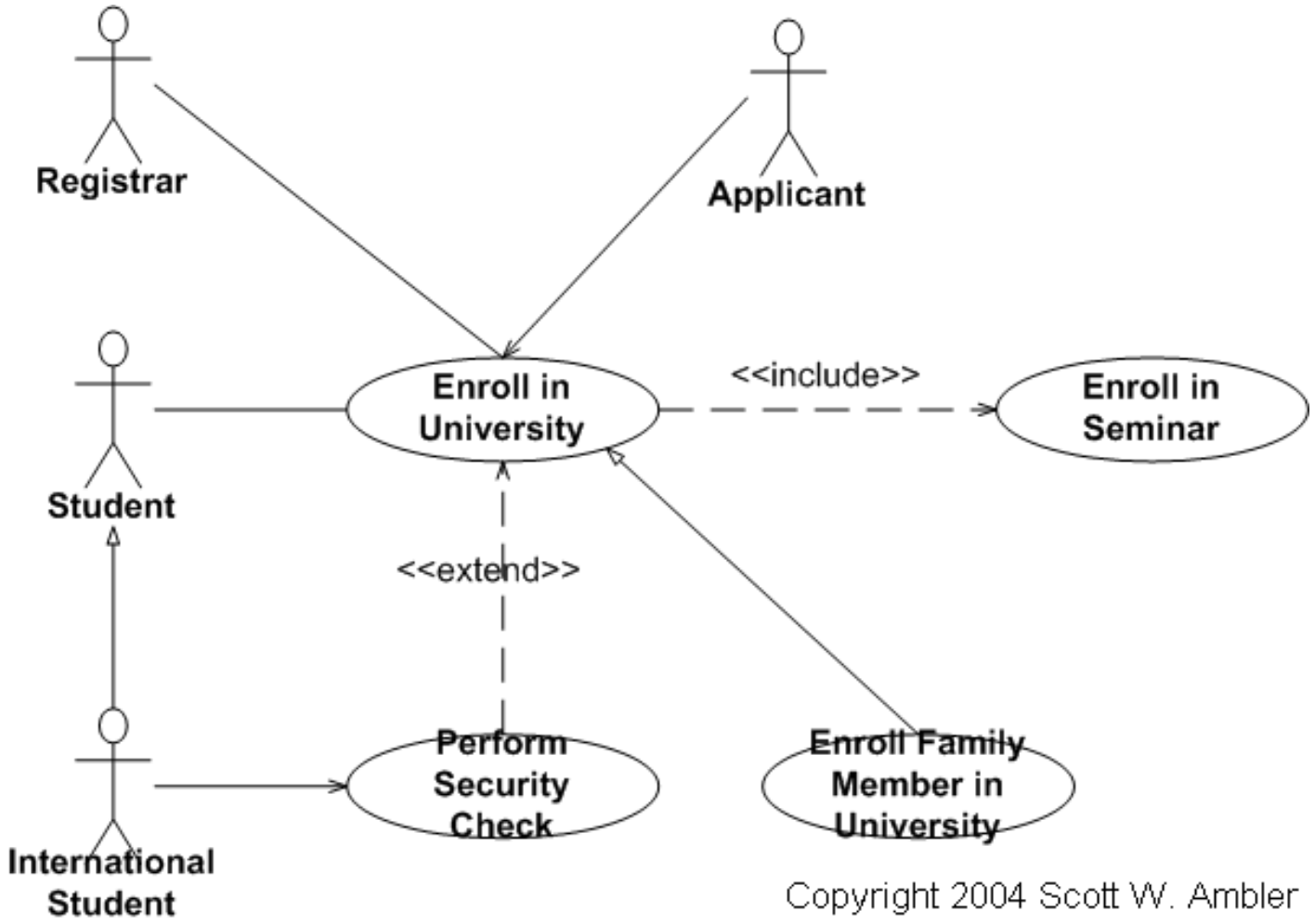


Extend with condition


Generalization

Use Case Diagram: Shapes

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Exercise: Group

1. Open “SAD-Tutorial-02-Cases.pdf” file.
2. Read the first case.
3. Create the Use Case Diagram for it on the whiteboard.

Exercise: Individual

1. Take 15 minutes to read the second case and draw its Use Case Diagram (on your own).

2. Anyone would like to draw their version on the whiteboard?

3. Discuss how your Use case diagram is similar or different from the one on the whiteboard.

Note: You can create it on a piece of paper or use any online tool.

The End

See you at the lecture.